

## TIP4PATLIBS — Course Overview

*Six modules · six 45-minute blocks · EPO Technology Intelligence Platform*

**What this document is.** The frame around the six module documents: what a participant can do at the end, how the modules depend on each other, how to run the material as a course, and what is deliberately not in it.

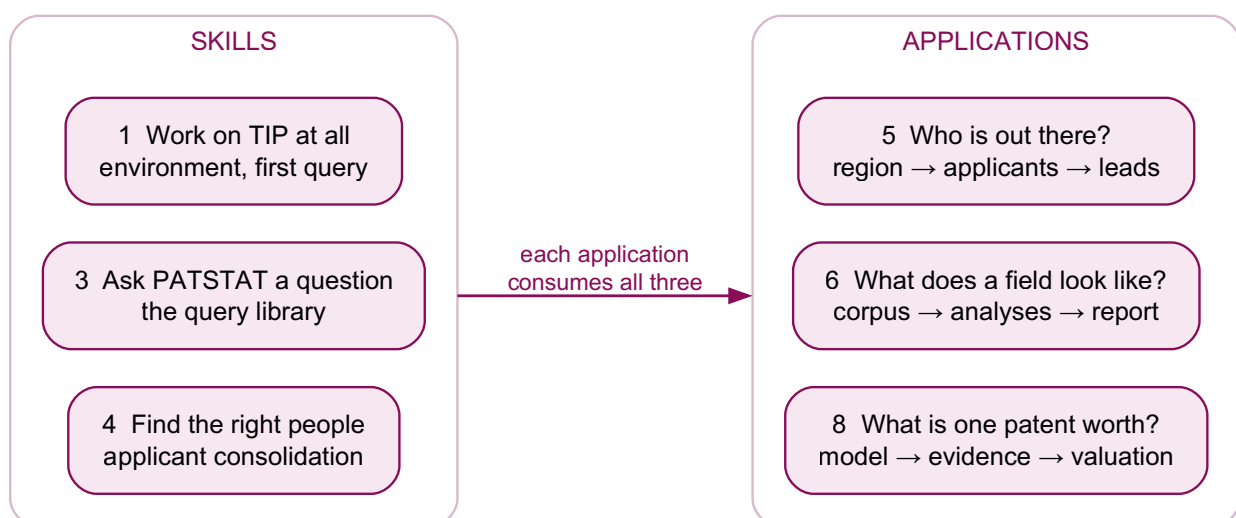
### The overall goal

A PATLIB adviser can take a real question from a real client — *"who is active in my region?", "what does this technology field look like?", "is this patent worth anything?"* — and answer it from PATSTAT on TIP, with a defensible method, a reproducible notebook and an artifact the client can keep. And they can say which parts of that answer are evidence and which are judgement.

That last clause is what separates this course from a tool demonstration. Every module returns to it, and module 5 makes it the output.

### The chain — three skills and three applications

The six modules are not six topics. They are **three skills** and **three applications**, and each application consumes all three skills.



| # | Module                            | Assumes | Adds                          | One-sentence goal                                                                                    |
|---|-----------------------------------|---------|-------------------------------|------------------------------------------------------------------------------------------------------|
| 1 | Start with TIP                    | —       | the working environment       | <i>I can open TIP, connect to PATSTAT, write my first query, and keep my setup across a restart.</i> |
| 3 | The Query Library                 | 1       | the question repertoire       | <i>I can pick a ready-made query, adapt its parameters, and read the result.</i>                     |
| 4 | PATSTAT Explorer                  | 1, 3    | applicant identity            | <i>I know why one company is forty different names, and I can consolidate them.</i>                  |
| 5 | Regional Lead Generation          | 1, 3, 4 | a client-facing output        | <i>I can profile a region's applicants and segment them into lead tiers.</i>                         |
| 6 | Patent Landscape Reports          | 1, 3, 4 | a publishable analysis        | <i>I can define a corpus with a defensible search strategy and turn it into a report.</i>            |
| 8 | IPScore — what is a patent worth? | 1, 3, 4 | the evidence/ judgement split | <i>I can value a patent, and state exactly how much of that number is checkable.</i>                 |

**Modules 4, 5 and 6 are parallel, not sequential.** A PATLIB that never does valuations can stop after module 4 and has lost nothing. Say this to participants — it lowers the barrier, and it is true.

**Why module 5 sits last.** Not because it is hardest, but because it is where the course's recurring theme becomes the deliverable. Module 1 says *an analysis you cannot open up is one you cannot defend*. Module 2 adds that some of what you open up is a decision you made. Module 3 makes you state your coverage limit. Module 4 makes you state your exclusions. Module 5 prints the split on the front page of the report.

## Skipped module numbers

The numbering follows the repository folders, so two numbers are missing here on purpose:

|   |                                                                                                                                                      |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2 | 9_misc/legacy/ — earlier worked end-to-end examples, superseded by modules 3–5                                                                       |
| 7 | 9_misc/ipscore/ — Riccardo Priore's workshop-ready IPScore tools. Not taught as a block here; module 5 is the course's own rebuild of the same model |

## How to run the material

### As a course — six 45-minute blocks

Each module document is one block, timed the same way:

| Phase                   | Time     | What happens                                                                                       |
|-------------------------|----------|----------------------------------------------------------------------------------------------------|
| <b>Introduction</b>     | ≈ 7 min  | The client question this module answers, and why the obvious approach fails                        |
| <b>Working through</b>  | ≈ 28 min | The notebooks, step by step, with the traps named before they are hit                              |
| <b>Learning outcome</b> | ≈ 10 min | What now exists, four self-check questions, and a transfer exercise for the participant's own desk |

Roughly 12–15 minutes of the working-through phase is reading; the rest is running cells and looking at results.

**A note on module 5.** Its full material is 8,272 markdown words across four notebooks — more prose than modules 1, 2, 3 and 5 combined. `05_ipscore.md` therefore covers a **core path**: the model plus a guided read of the finished report. The implementation chain is a second block, `05_ipscore_part2.md`. Nothing is missing from the argument; only the code walk is deferred.

### As a workshop — two 45-minute blocks

For a showcase there is a slide version of the same material, three slides per module:

| Block                   | Modules | Slides | Per module |
|-------------------------|---------|--------|------------|
| <b>A — Skills</b>       | 1, 3, 4 | 9      | 15 min     |
| <b>B — Applications</b> | 5, 6, 8 | 9      | 15 min     |

Plus the chain diagram above as an opening slide and one closing slide — **20 slides**. The slides are generated *from* these documents, so there is one source of truth.

### As a handout

Each document is dual-track: **the running text addresses the participant**, and material for whoever is running the session sits in boxes labelled **Trainer**. Traps sit in boxes labelled **Trap**. One file serves as handout and as script, which is what makes the material reusable by another PATLIB.

## Prerequisites for the whole course

- A **TIP account** with JupyterLab access. Nothing is installed locally.
- **No Python and no SQL** are assumed. Both appear at the level needed and no further.
- Modules 1, 2, 3 and 4 need PATSTAT PROD; one cell in module 4 additionally uses BigQuery and ships with its output.

**Data edition throughout: PATSTAT Global, Autumn 2025.** Every number in the material depends on it — module 3 opens by checking which edition you are actually on, and that habit is worth keeping.

## What ships executed, and what does not

This matters for reading ahead, and it is not an accident.

| Module             | Executed cells | Why                                                                       |
|--------------------|----------------|---------------------------------------------------------------------------|
| 1 Start with TIP   | 0 of 25        | You are meant to run it — it configures <i>your</i> environment           |
| 2 Query Library    | 6 of 10        | Partly interactive; the browser has to be run                             |
| 3 PATSTAT Explorer | 7 of 7         | Read it before you run it                                                 |
| 4 Lead Generation  | 8 of 31        | You are meant to run it for <b>your own</b> region                        |
| 5 Patent Reports   | 29 of 29       | Read as a <b>finished report</b> — the outputs <i>are</i> the deliverable |
| 6 IPScore Rebuild  | 37 of 38       | Read as a finished valuation                                              |

**Trap.** Modules 5 and 6 must **never** be re-run to "tidy" their outputs. For module 4 in particular the committed outputs are contributed material.

## Credits

- Modules 1, 2, 3, 4 and 6 — **Arne Krüger**, mtc.berlin / depa.tech.
- Module 4 — landscape analyses after **Riccardo Priore**, Centro PATLIB, AREA Science Park, reworked to match this course's look. The IPScore reference, outside this block, is his material as well.
- **IPScore is an EPO tool.** Module 5 is this course's own implementation of the IPScore 3.01 model, verified against the EPO's own workbook.

License: **EPO Internal Use.**

## The six documents

|                        |                                                                |
|------------------------|----------------------------------------------------------------|
| 01_start-with-tip.md   | The environment, and your first query written by hand          |
| 02_query-library.md    | Choosing from queries somebody already made defensible         |
| 03_patstat-explorer.md | Why one company is forty names, and what to do about it        |
| 04_lead-generation.md  | A region's applicants, ranked and segmented into lead tiers    |
| 05_patent-reports.md   | A corpus, a battery of analyses, and one self-contained report |
| 05_ipscore.md          | A valuation, and how much of it is evidence                    |

The planning document behind them — including the module-8 restructuring argument and the rendering plan — is `../plan-course-material.md`.